NFL Two-Point Conversion Analysis

1. Project Overview

This project analyzes over 1,700 NFL two-point conversion attempts across passing and rushing plays. The goal is to uncover insights into offensive play-calling tendencies, conversion success rates, player involvement, and positional performance to support data-driven decision-making and strategic game planning.

2. Dataset Summary

Dataset: twoPtPass.csv
- Rows: 1,217 - Columns: 9 - Key Features:
- Id (Pass, Play, Team, Player)
- Pass Position
- Sack, Completion, Null
- Conversion (Good/Fail)
Dataset: twoPtRush.csv
- Rows: 435 - Columns: 7 - Key Features:
- Id (Pass, Play, Team, Player)
- Rush Position
- Result (Good/Fail)
- Null
Dataset: twoPtRec.csv
- Rows: 1,153 - Columns: 8 - Key Features:
- Id (Pass, Play, Team, Player)
- Receiving Position
- Completion, Null
- Result (Good/Fail)

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported dataset using `pandas`.

- **Initial Exploration:** Used `df.head()` to check structure.

[20]:

	twoPtRecId	playId	teamId	playerId	twoPointRecPosition	twoPtRecResult	twoPtRecComp	twoPtRecNull
0	410001	115	3800	19900017	RB	failed	0.0	0
1	410002	2564	3300	19960135	WR	good	1.0	0
2	410003	3999	325	20040199	WR	good	1.0	0
3	410004	4990	3900	20000008	WR	failed	0.0	0
4	410005	7442	750	20030076	TE	good	1.0	0

- **Data Cleaning:** Removed unnecessary columns, including null placeholder fields and redundant identifiers (teamId, playerId, and twoPtRecId), to prepare them for analysis.

- **Dataset Integration:** Merged passing and receiving datasets using the shared playId to accurately associate quarterbacks with their intended receivers for each two-point conversion attempt.

[6]:

	playId	twoPtPassPosition	twoPtPassConv	twoPtPassSack	twoPtPassComp	twoPointRecPosition
0	115	QB	failed	0	0	RB
1	2564	QB	good	0	1	WR
2	3999	QB	good	0	1	WR
3	4990	QB	failed	0	0	WR
4	7442	QB	good	0	1	TE
...	...	...	...	...	...	...
1212	15079041	QB	good	0	1	WR
1213	15079221	QB	good	0	1	WR
1214	15082774	QB	failed	0	0	WR
1215	15083215	QB	good	0	1	WR
1216	15084200	QB	failed	0	1	WR

- **Feature Engineering:**
  - Created target_status column by analyzing twoPtPassSack and dftwoPtPassComp.

- **Data Consistency Check:** Verified if `twoPtPassSack` and `twoPtPasscomp` were redundant; dropped `twoPtPassSack` and `twoPtPasscomp`.

- **Database Integration:** Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL

We performed structured analysis in PostgreSQL to answer key questions:

1. **2-Point Attempts by Play Type** – Compared the total number of two-point conversion attempts between passing and rushing plays.

	play_type text	total_attempts bigint
1	Passing	1217
2	Rushing	435

2. **Success Rate by Play Type** – Analyzed and compared the conversion success rates of passing versus rushing two-point attempts.

	play_type text	total_attempts bigint	successful_attempts bigint	success_rate_percent numeric
1	Passing	1217	520	42.73
2	Rushing	435	255	58.62

3. **Most Targeted Receivers** – Identified the receivers who were targeted most frequently on two-point conversion pass attempts.

	targeted_position text	total_targets bigint
1	WR	751
2	TE	256
3	RB	114
4	FB	12
5	DE	4

4. **Receiver Conversion Efficiency** – Determined which receivers achieved the highest success rate on two-point conversion targets.

	targeted_position text	total_targets bigint	successful_targets bigint	success_rate_percent numeric
1	FB	12	7	58.33
2	TE	256	129	50.39
3	RB	114	55	48.25
4	WR	751	321	42.74

5. **Most Active Rushers** – Determined which rushing success rates to identify the most effective two-point conversion rushers.

	rusher_position text	rushing_attempts bigint
1	RB	277
2	QB	115
3	P	14
4	FB	8
5	WR	7
6	HB	7

6. **Rusher Conversion Efficiency** – Compared the rushing success rates to identify the most effective two-point conversion rushers.

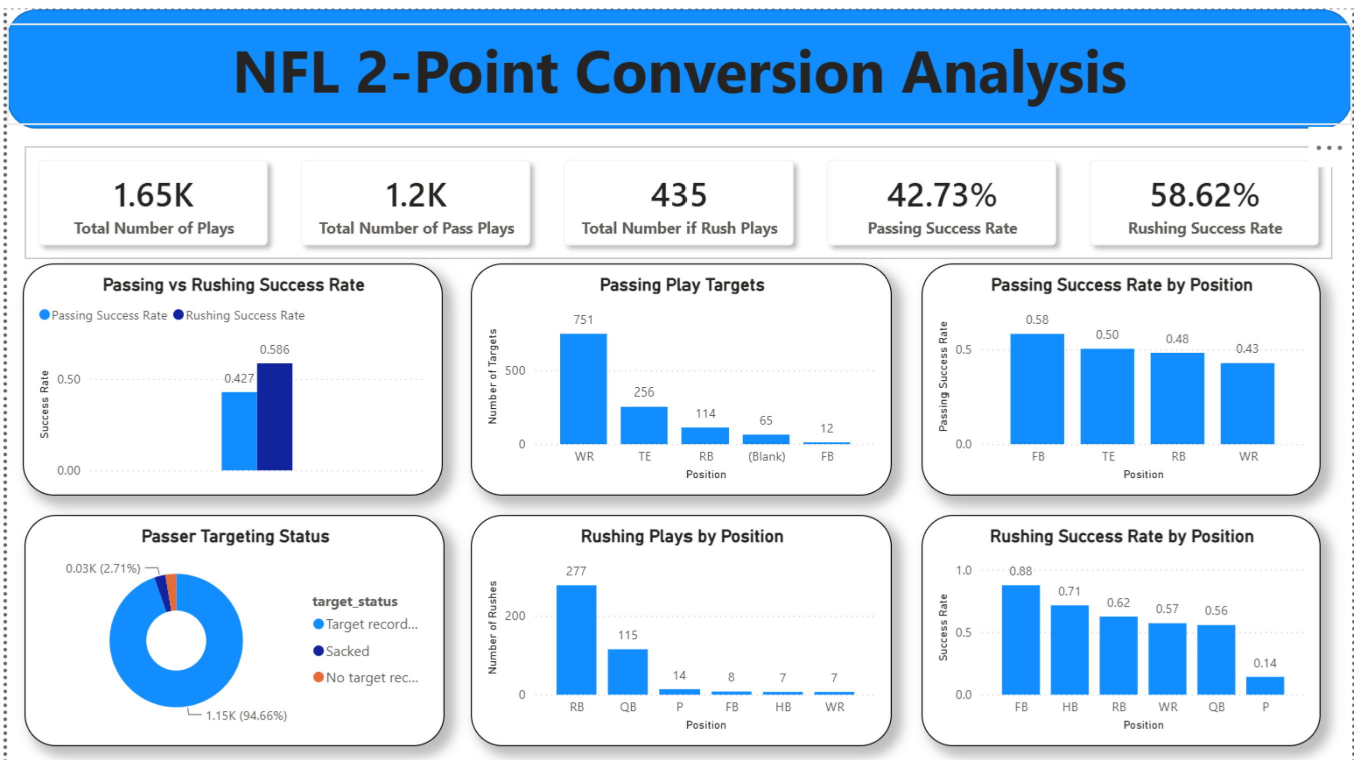
	rusher_position text	total_rush_attempts bigint	successful_rushes bigint	rushing_conversion_rate_percent numeric
1	FB	8	7	87.50
2	HB	7	5	71.43
3	RB	277	173	62.45
4	WR	7	4	57.14
5	QB	115	64	55.65
6	P	14	2	14.29

7. **Most Involved Players** – Identified the players who participated in the greatest number of two-point conversion attempts across both passing and rushing plays.

	player_position text	total_attempts_involved bigint
1	WR	758
2	RB	391
3	TE	257
4	QB	118
5	FB	20
6	P	14

5. Dashboard in Power BI

Finally, we built a dashboard in **Power BI** to present insights visually.



6. Recommendations

- **Increase the Use of Rushing Plays** – Since rushing two-point conversions achieved a 58.62% success rate, compared to 42.73% for passing plays, teams should consider favoring rushing attempts in high-leverage situations where maximizing the probability of conversion is the priority.

- **Expand the Use of Tight Ends** - Tight ends demonstrated the highest passing conversion success rate (96.99%), thus increasing the number of designed plays targeting tight ends could improve overall passing efficiency while reducing offensive predictability.

- **Reduce Overreliance on Wide Receivers** – Wide receivers accounted for 751 total targets, significantly more than any other position. Diversifying passing targets by utilizing tight ends and running backs may create a more balanced offense and make defenses less able to anticipate play calls.

- **Evaluate Underperforming Passing Concepts** - Despite strong success rates among wide receivers and tight ends, the overall passing conversion rate remained lower because of unsuccessful attempts involving less effective player combinations. Review these lower-performing play designs and refine or replace them with more efficient alternatives.